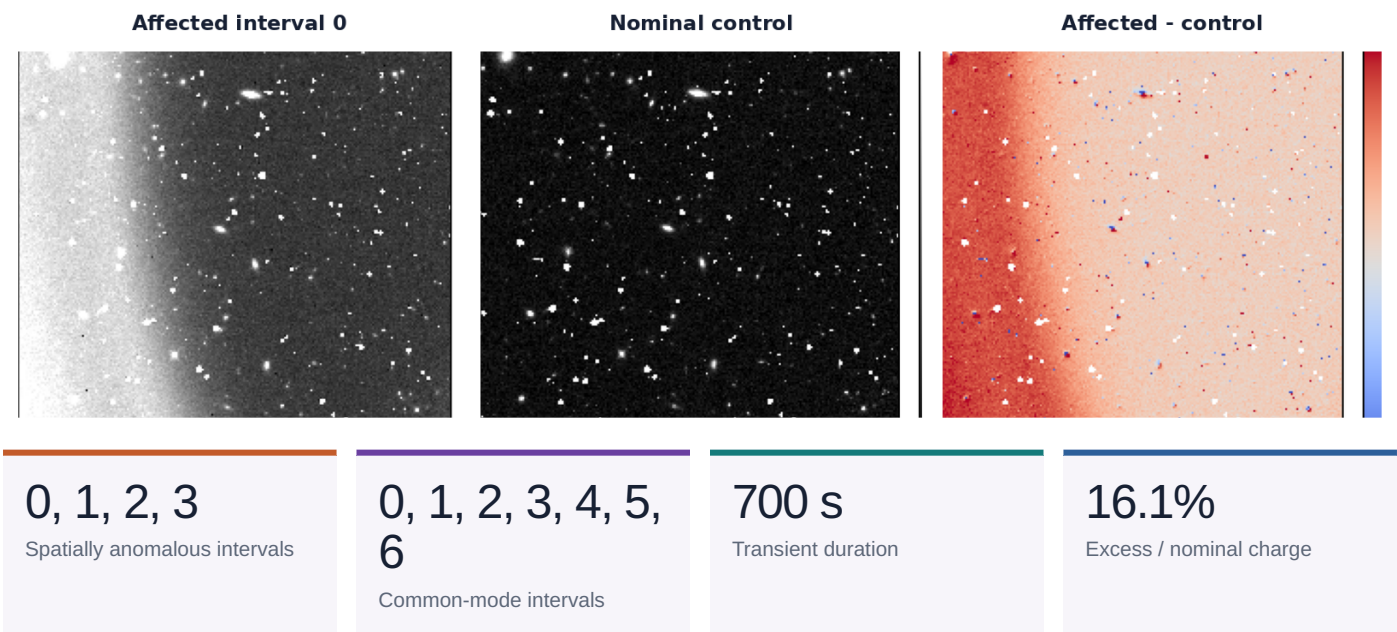


Read-level diagnosis of time-variable contamination in a HgCdTe focal plane array

A reproducible WFC3/IR analysis showing why spatial inspection alone can miss a lower-amplitude transient tail - and how read selection changes the final count-rate product.



The visible spatial gradient ends before the temporal contamination. A spatial-only diagnostic would terminate the anomaly too early and retain intervals with measurable control-relative excess.

Analysis by Brooks Photonics using public HST WFC3/IR measurements from MAST.
No client data are represented.

July 2026

EXECUTIVE SUMMARY

The result is a measurement-condition diagnosis, not a detector-material diagnosis

The affected exposure contains two physically distinct regimes. Intervals **0, 1, 2, 3** show pronounced left-right nonuniformity. A lower-amplitude detector-wide excess continues through interval **6**, giving a total transient duration of **700.002 s**.

Most consistent mechanism: time-variable external background

The effect decays during one exposure, is absent from the matched control, survives source masking, and changes the reconstructed slope.

- 1

Reconstruct charge

Convert calibrated count-rate reads into cumulative charge.
- 2

Measure intervals

Calculate consecutive-read rates instead of trusting one final slope.
- 3

Separate modes

Quantify spatial asymmetry and common-mode excess independently.
- 4

Refit the ramp

Compare all-read and post-transient slopes with the archived FLT.

53.85%

Peak affected asymmetry

Compared with 1.83% in the nominal control.

0.4958

Peak excess rate

e-/s/pixel after matched-control and late-offset correction.

94.25%

Source-free pixels retained

Classification agreement remains 100%.

2,000

Bootstrap replicates

Every run retains the nominal 700 s classification duration.

DECISION METRIC	RESULT
Integrated full-frame excess	197.26 e-/pixel
Robust tile-median estimate	150.03 e-/pixel
Bootstrap 95% interval	144.61-159.50
Pixels ever carrying DATAREJECT	35.80%

01 / DATA AND METHOD

The analysis preserves read-level evidence that the pipeline itself can reject

Four public products are used: two calibrated IMA ramps and their archived FLT count-rate products. The affected and nominal exposures come from the same HST program and visit, providing a defensible matched comparison.

Interval-rate construction

For calibrated read rate R_i at time t_i , cumulative charge is $Q_i = R_i t_i$. Consecutive-read rate is then $(Q_i - Q_{i-1}) / (t_i - t_{i-1})$.

Data-quality policy

Every standard 16-bit WFC3 DQ flag is rejected except **DATAREJECT = 8192**. That bit is retained because it is generated by ramp fitting and is diagnostically relevant, not a permanent material defect.

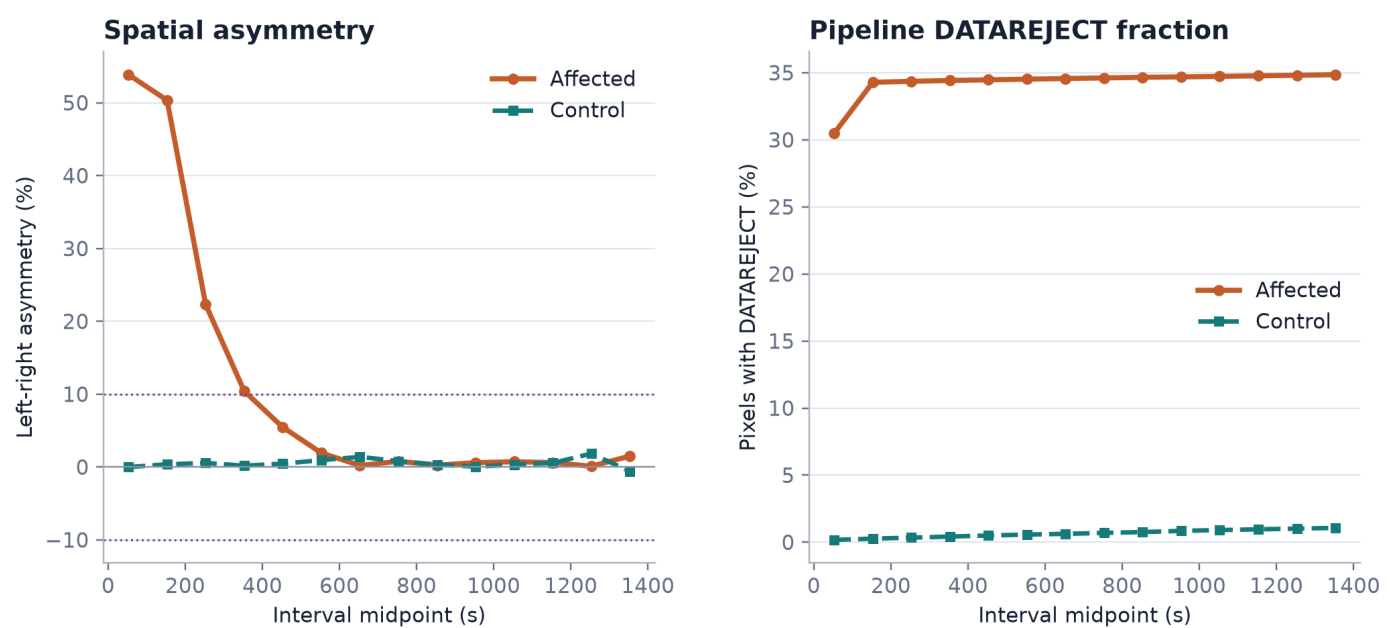


Figure 1. The early spatial asymmetry and the pipeline DATAREJECT response evolve together. Color, line style, and marker shape redundantly identify affected and control exposures.

INPUT PRODUCT	ROLE
icqtbbsbxq_ima.fits	Affected calibrated nondestructive reads
icqtbbsbc0q_ima.fits	Nominal matched-control reads
icqtbbsbxqflt.fits	Archived affected ramp-fit comparison
icqtbbsbc0qflt.fits	Archived nominal ramp-fit comparison

02 / SPATIAL EVOLUTION

The strong gradient collapses first; the common-mode tail persists

Representative interval maps show the transition from an obvious left-side enhancement to a detector-wide residual that is difficult to identify from spatial appearance alone.

Representative WFC3/IR instantaneous interval-rate maps

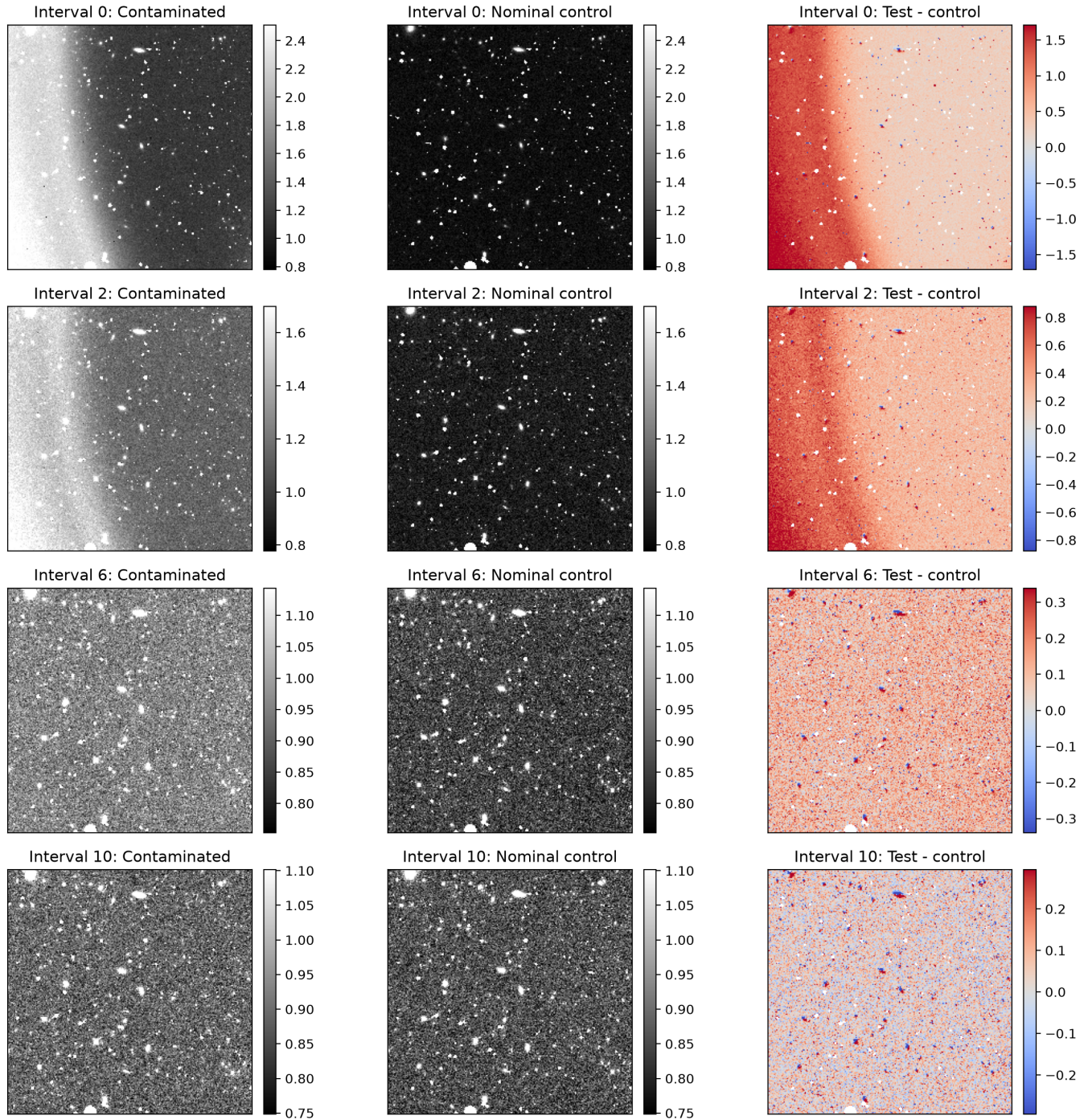


Figure 2. Affected exposure, nominal control, and matched difference at intervals 0, 2, 6, and 10. The color scales are interval-specific so spatial structure remains visible as the transient decays.

03 / TEMPORAL DIAGNOSIS

A spatial-only criterion would stop two intervals too early

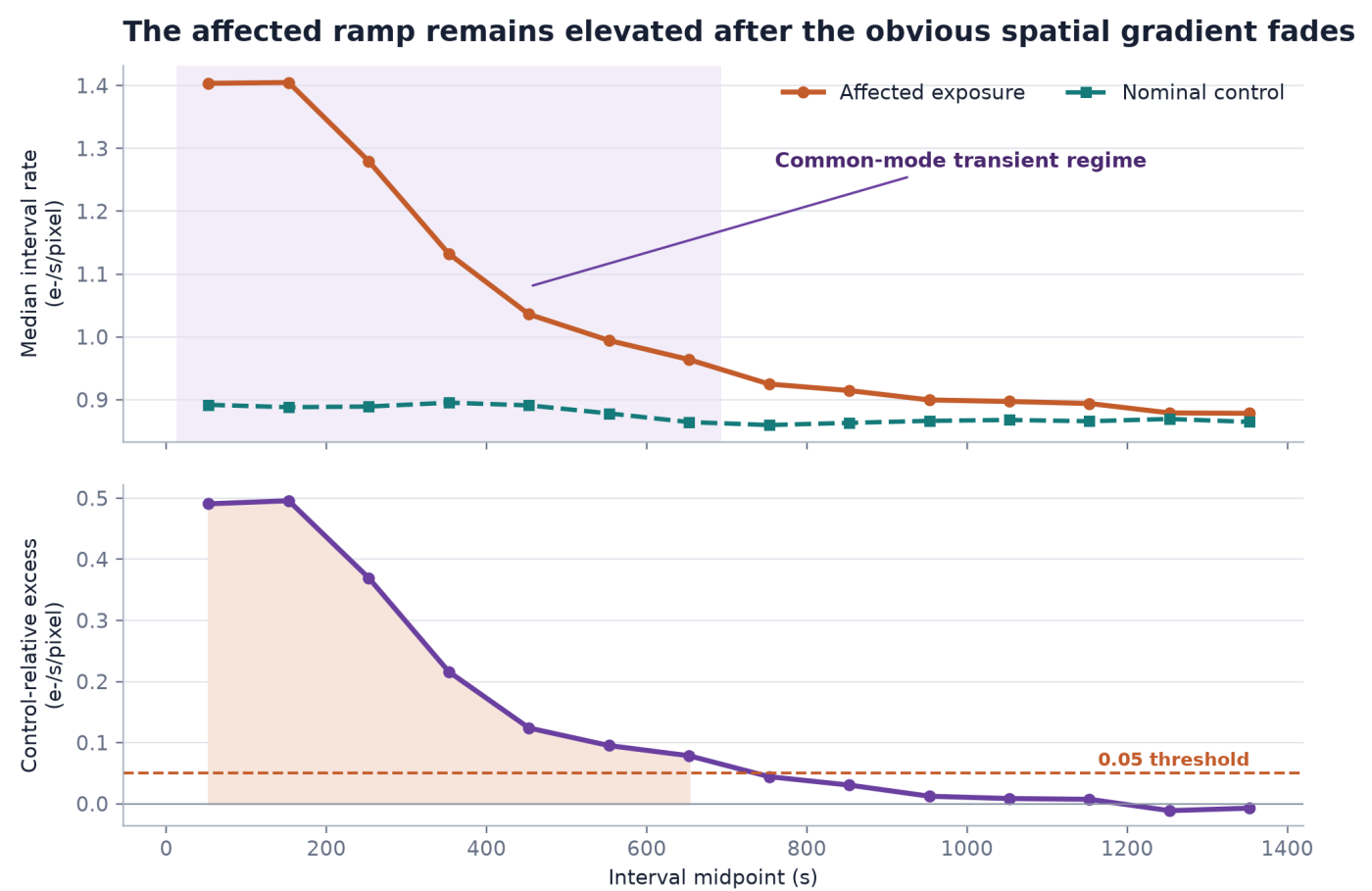


Figure 3. The shaded region marks intervals 0-6, which exceed the nominal 0.05 e-/s/pixel common-mode threshold after control subtraction and late-time offset removal.

0-3

Spatial regime

Strong left-right structure is independently visible.

4-6

Hidden tail

Common-mode excess remains after spatial asymmetry subsides.

0.02096

Static offset removed

e-/s/pixel estimated from the late ramp.

Engineering implication: read acceptance should not be based on map uniformity alone. Temporal controls are necessary even after the detector image appears spatially stable.

04 / ROBUSTNESS

The conclusion survives threshold, baseline, source-mask, and spatial-resampling tests

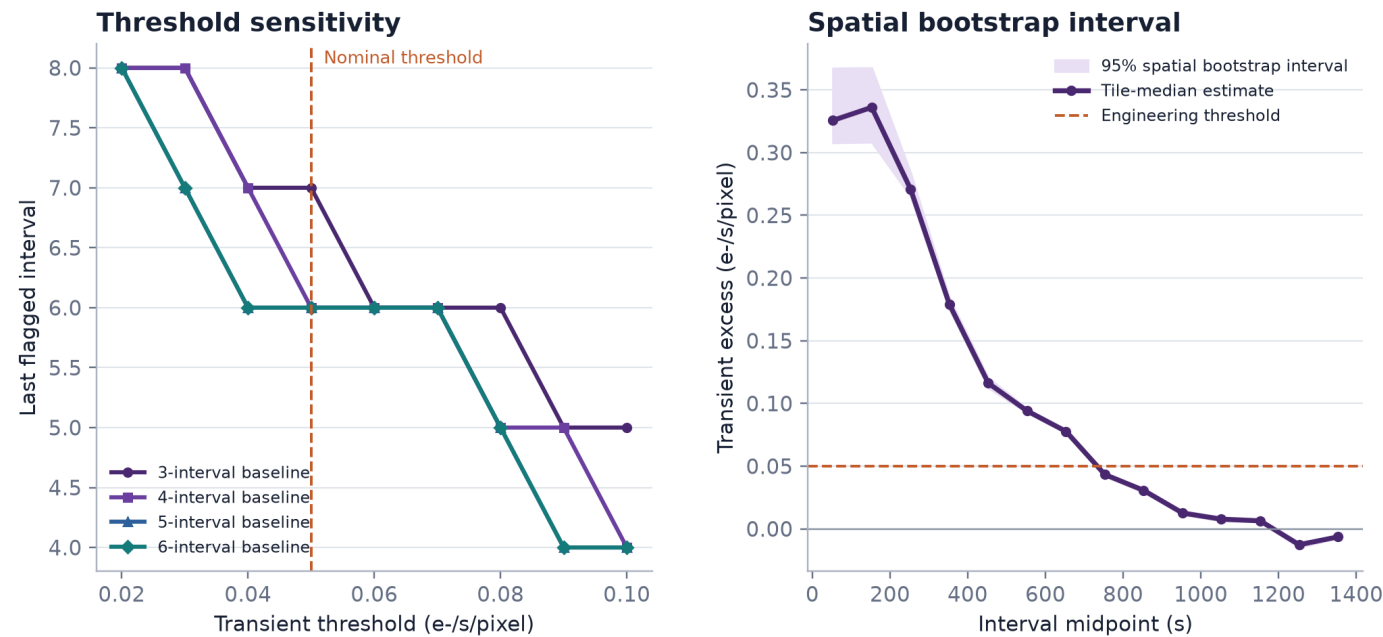


Figure 4. Threshold/baseline sensitivity and spatial block-bootstrap uncertainty. The nominal classification is not an isolated tuning outcome.

Source-free validation

The late-control source mask retains **94.25%** of interior pixels. Classification agreement is **100%**, and the maximum change in interval excess is only **0.00313 e-/s/pixel**.

Spatial bootstrap

779 independent 32 x 32 pixel tiles are resampled **2,000** times. The 95% interval on integrated excess is **144.61-159.50 e-/pixel**.

Estimator note: the full-frame median and spatial tile-median answer slightly different questions. The tile estimator deliberately reduces the leverage of strongly illuminated spatial regions and is used for the robust uncertainty interval.

05 / INDEPENDENT RECONSTRUCTION

Removing the transient regime materially changes the inferred count-rate product

A free-intercept, DQ-aware ordinary-least-squares slope is fit to cumulative charge. The post-transient fit begins at **702.934 s**, after interval 6.

Archived and independently reconstructed count-rate products

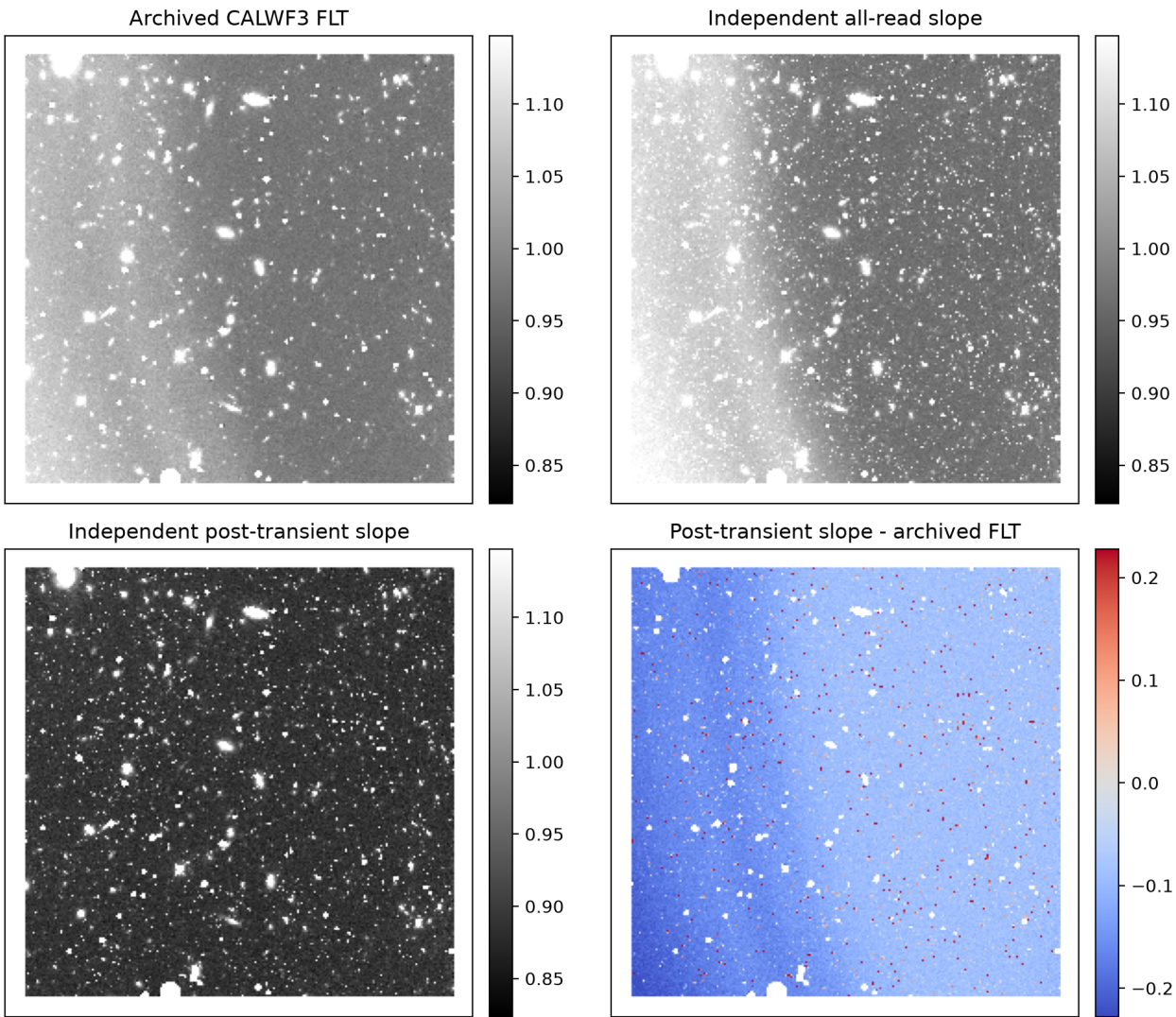


Figure 5. Archived CALWF3 FLT, independent all-read slope, post-transient slope, and post-transient minus archived residual.

PRODUCT OR FIT	SOURCE-FREE MEDIAN	LEFT-RIGHT ASYMMETRY
Archived CALWF3 FLT	1.00296 e-/s/pixel	6.32%
Independent all-read fit	0.99992 e-/s/pixel	9.55%
Independent post-transient fit	0.89337 e-/s/pixel	0.40%

Recommended acceptance rule: evaluate temporal and spatial integrity separately

The final fitted product retains measurable influence from the time-variable ramp history. The post-transient fit is 0.11108 e-/s/pixel below the archived FLT median and reduces left-right asymmetry from 6.32% to 0.40%.

- 1

Inspect consecutive-read maps
Do not accept a final slope product without reviewing interval-level spatial behavior.
- 2

Classify spatial and common-mode effects independently
A detector can appear spatially stable while retaining a measurable temporal background.
- 3

Use a matched control or validated late-ramp baseline
Separate a static exposure offset from a decaying transient before thresholding.
- 4

Quantify decision sensitivity
Report how the classification changes across plausible thresholds and baseline windows.
- 5

Refit after excluding the transient regime
Compare the reconstructed slope and residual structure against the archived or production result.

Mechanism assessment

HYPOTHESIS	ASSESSMENT
Time-variable external background	Most consistent Temporal decay, spatial evolution, and matched-control result agree.
Stable gain or flat-field nonuniformity	Disfavored; expected to persist and appear similarly in the control.
Intrinsic dark-current nonuniformity	Strongly disfavored; the detector-wide pattern decays within one exposure.
Readout-quadrant offset	Secondary contribution possible, but insufficient to explain the smooth decay.
Isolated cosmic-ray event	Cannot explain detector-wide, smoothly decaying background behavior.

Interpretation boundary. This case does not infer HgCdTe composition, carrier lifetime, dark-current mechanism, responsivity, detectivity, or intrinsic material quality.

APPENDIX / REPRODUCIBILITY AND SOURCES

The complete case is reproducible from pinned public products

Reproducibility controls

Exact MAST product URIs, byte counts, and SHA-256 digests are stored in the repository inventory. The workflow downloads all four products, verifies them, regenerates the analysis, runs unit and lint checks, and publishes the PDF and derived data as a GitHub Actions artifact.

- Public data
- Pinned provenance
- Automated verification

Primary sources

- STScI.** WFC3/IR IMA Visualization Tools with an Example of Time Variable Background.
- STScI.** Correcting for Scattered Light in WFC3/IR Exposures: Manually Subtracting Bad Reads.
- STScI.** WFC3 Data Handbook, file structure and IR calibration sections.
- MAST.** Public HST product archive and download API.

Design and communication standard

This report uses an insight-led title hierarchy, a consistent spacing scale, direct chart annotations, and redundant visual encodings so color supports the analysis without carrying meaning alone.

OUTPUT	PURPOSE
case_summary.json	Top-level quantified findings
exposure_comparison.csv	Interval-level matched comparison
threshold_sensitivity.csv	Decision robustness sweep
bootstrap_interval_uncertainty.csv	Spatial uncertainty intervals
flt_reconstruction_metrics.json	Archived versus independent fit comparison

Brooks Photonics

Independent, physics-based analysis of infrared detector electrical, spectral, temporal, and noise data.